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SCENARIO BASED

Course Code: DSA 0308

Course Name: Natural Language Processing

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Q1. Morphological Parsing and Normalization

1. Problem Statement

Document classification and search engines frequently encounter different word forms that are semantically related but differ in their morphological structure. For example, the words activate, activation, and reactivation are related to the concept of making something active, but they represent different grammatical forms and meanings.

If these words are indexed independently, a search system may fail to group relevant documents consistently. Therefore, a morphological processing system is required to identify the underlying lexical relationship, recognize derivational prefixes and suffixes, preserve meaningful distinctions, and normalize related terms to a common representation.

The proposed system uses rule-based morphological analysis to identify the structure of each word and generate a structured report suitable for semantic indexing and document classification.

2. Objectives

1. To identify the prefixes, roots, and suffixes in the given input words.
2. To determine the sequence of derivational transformations and the resulting changes in word class.
3. To interpret the semantic effect of each morphological modification.
4. To normalize related word forms to the common lexical representation activate.
5. To generate a structured report that supports consistent document classification and information retrieval.

3. Theoretical Background

(i) Morphology

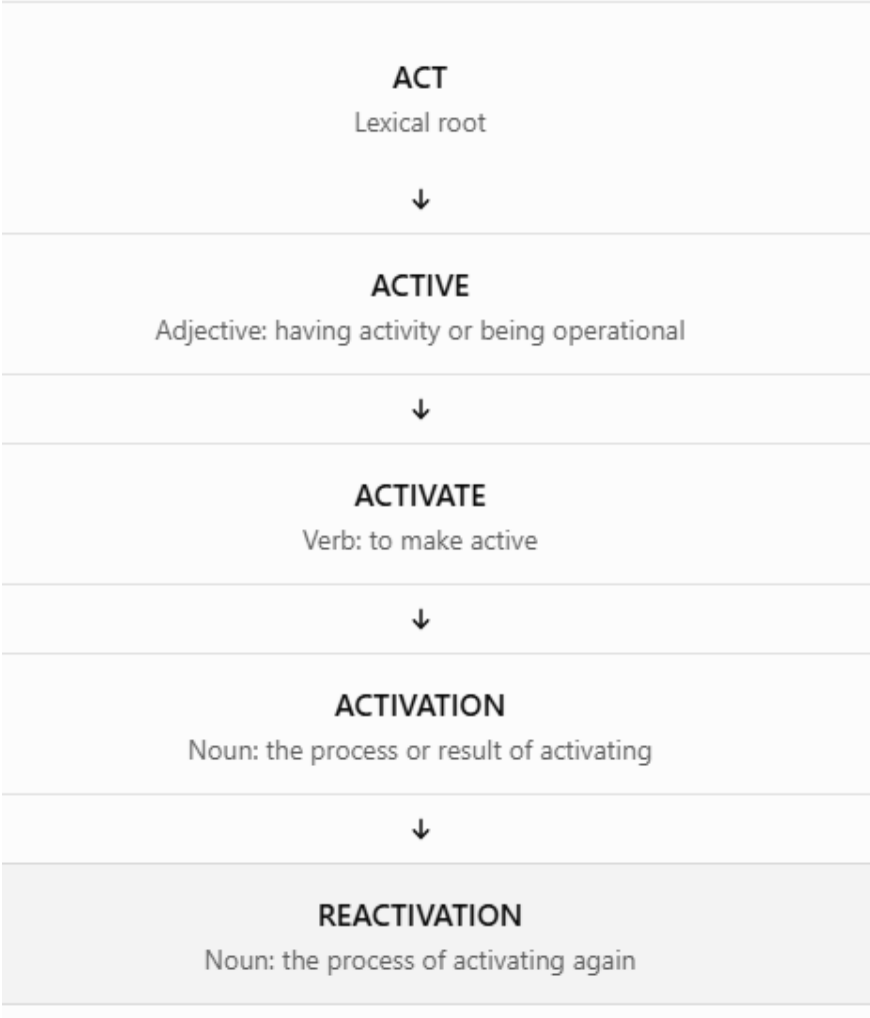
Morphology is the branch of Natural Language Processing (NLP) that studies the internal structure of words and the meaningful units from which they are formed. These units are called morphemes.

The main morphological components are:

- Root: The fundamental lexical element that carries the central meaning.
- Prefix: A morpheme added before a base to modify its meaning.
- Suffix: A morpheme added after a base to modify meaning or grammatical category.
- Derivational morphology: The process of forming a new word or lexical meaning by adding derivational affixes.
- Normalization: The process of mapping related word forms to a consistent representation for indexing and retrieval.

(ii) Derivational Structure

The three input words are related through the following derivational sequence:



The derivational relationship is represented as:

- active + -ate → activate
- activate + -ion → activation
- re- + activate → reactivate
- reactivate + -ion → reactivation

The prefix re- introduces the meaning of repetition, whereas the suffix -ion forms a noun associated with an action or process.

4. Morphological Analysis of the Input Words

Original Word	Prefix	Root/Base	Suffix	Word Class	Semantic Interpretation
activate	None	active	-ate	Verb	To make something active
activation	None	activate	-ion	Noun	The process or result activating
reactivation	re-	activate	-ion	Noun	The process of activating again

Interpretation: The lexical root act provides the central semantic relationship. The intermediate base active leads to the verb activate. The suffix -ion converts the verb into a noun, while the prefix re- indicates that an action is performed again.

For accurate morphological analysis, the root act and the immediate base activate must be distinguished. For example, reactivation is best analyzed as re- + activation at the immediate word-formation level, while its deeper lexical relationship can be traced to act.

5. Algorithm

1. Start the program.
2. Initialize the input words: activate, activation, and reactivation.
3. Create a rule-based morphological dictionary containing the prefix, root, suffix, derivational sequence, word class, and semantic interpretation of each word.
4. Read each input word and identify its morphological components using the defined rules.
5. Determine the derivational sequence that explains how the word is formed.
6. Identify the grammatical word class and interpret the semantic effect of each prefix or suffix.
7. Normalize each input word to the lexical representation activate.

8. Display the original word, prefix, root, suffix, derivational sequence, word class, semantic interpretation, and normalized form.
9. Stop the program.

6. Python Program

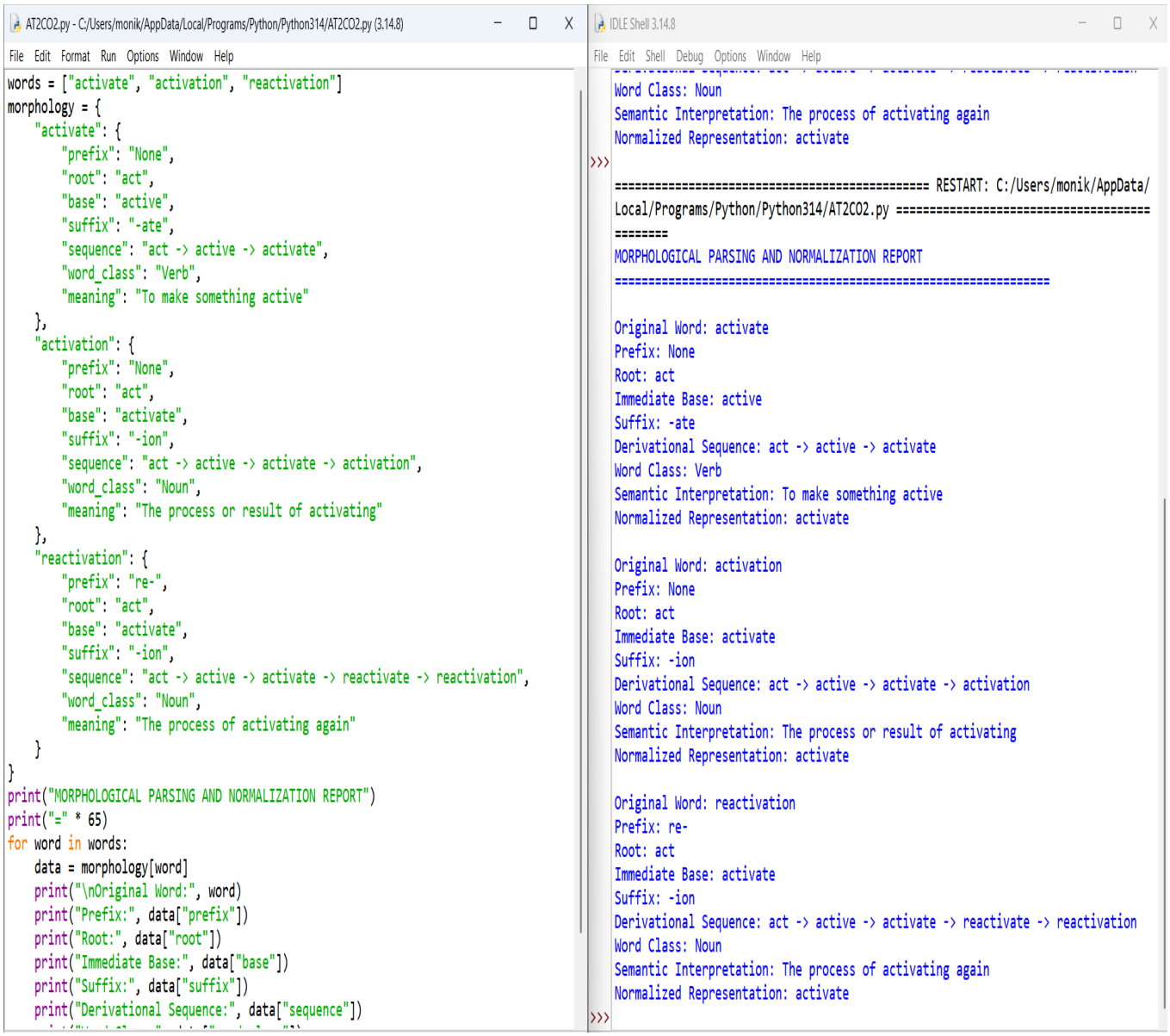
```
words = ["activate", "activation", "reactivation"]

morphology = { "activate": {
        "prefix": "None",
        "root": "act",
        "base": "active",
        "suffix": "-ate",
        "sequence": "act -> active -> activate",
        "word_class": "Verb",
        "meaning": "To make something active"
    },
    "activation": {
        "prefix": "None",
        "root": "act",
        "base": "activate",
        "suffix": "-ion",
        "sequence": "act -> active -> activate -> activation",
        "word_class": "Noun",
        "meaning": "The process or result of activating"
    },
```

```
    "reactivation": {
        "prefix": "re-",
        "root": "act",
        "base": "activate",
        "suffix": "-ion",
        "sequence": "act -> active -> activate -> reactivate -> reactivation",
        "word_class": "Noun",
        "meaning": "The process of activating again"
    }
}
print("MORPHOLOGICAL PARSING AND NORMALIZATION REPORT")
print("=" * 65)
for word in words:
    data = morphology[word]
    print("\nOriginal Word:", word)
    print("Prefix:", data["prefix"])
    print("Root:", data["root"])
    print("Immediate Base:", data["base"])
    print("Suffix:", data["suffix"])
    print("Derivational Sequence:", data["sequence"])
    print("Word Class:", data["word_class"])
    print("Semantic Interpretation:", data["meaning"])
```

print("Normalized Representation: activate")

7. Output



8. Result

The Python program successfully demonstrates morphological parsing and normalization of the three input words. It identifies their derivational relationships, word-class changes, and semantic differences, and maps them to a common normalized representation for document classification and semantic indexing.

Q2. Inflectional Morphology and Normalization

1. Problem Statement

Search optimization systems process text containing words that express the same lexical concept in different grammatical forms. For example, create, creates, and creating all refer to the action of creating, but they differ in their grammatical form and usage.

If these forms are indexed separately, a search engine may not retrieve all relevant documents for a query containing a different grammatical form. A normalization mechanism is therefore required to identify the base form of each word while preserving its underlying meaning.

The proposed system uses rule-based inflectional analysis to identify suffixes, classify grammatical features, handle spelling changes, and normalize the three input words to the common base form create.

2. Objectives

1. To identify inflectional suffixes in the input words.
2. To distinguish the base form, third-person singular present form, and present participle.
3. To apply spelling rules correctly when recovering the base form.
4. To normalize all three word forms to create without changing their underlying lexical meaning.
5. To generate a structured report suitable for search indexing and information retrieval.

3. Theoretical Background

(i) Inflectional Morphology

Inflectional morphology studies changes in a word's form that express grammatical information without creating a new dictionary word or fundamentally changing its lexical meaning.

For example:

- create is the base form of the verb.
- creates is the third-person singular present-tense form.
- creating is the present participle, used in progressive constructions such as "She is creating a document."

All three words share the same lexical base, although their grammatical functions differ.

(ii) Inflectional Suffixes

An inflectional suffix is added to a word to express a grammatical feature such as tense, agreement, number, or aspect-related usage.

In this scenario:

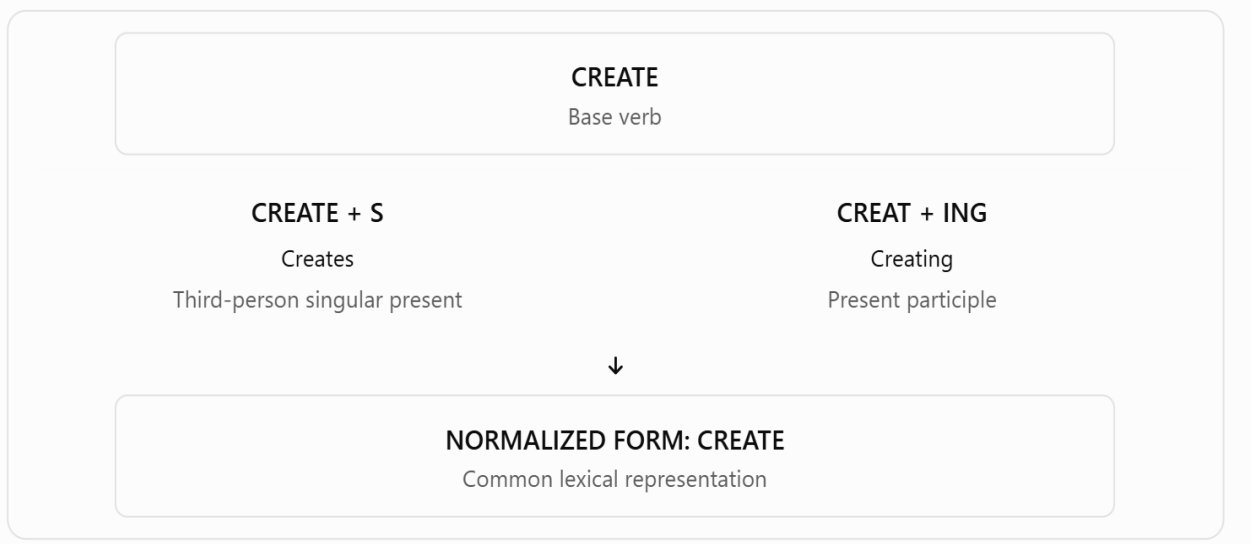
- -s marks third-person singular present-tense agreement in creates.
- -ing forms the present participle in creating.

The spelling rules must also be considered. The word create loses its final silent e before -ing is added:

create → creat + ing → creating

During normalization, the system restores the final e to recover create.

(iii) Morphological Structure



4. Morphological Analysis of the Input Words

Original Word	Identified Suffix	Morphological Structure	Grammatical Category	Normalized Form
create	None	create	Base form	create
creates	-s	create + -s	Third-person singular presen	create
creating	-ing	create - e + -ing	Present participle	create

Interpretation:

- Create: The base verb; no inflectional suffix is present.
- Creates: The suffix -s marks third-person singular present-tense agreement.
- Creating: The suffix -ing forms the present participle. In creating, the final silent e is dropped before the suffix is attached.

The normalized form is create for all three words because they share the same underlying lexical verb. Their grammatical differences are recorded separately.

5. Algorithm

1. Start the program.
2. Initialize the input list with create, creates, and creating.
3. Examine each word to determine whether it is the base form or contains an inflectional suffix.
4. If the word is create, classify it as the base form.
5. If the word ends in s and matches creates, identify -s and classify it as third-person singular present.
6. If the word matches creating, identify -ing and apply the spelling rule that restores the final e during normalization.
7. Extract the base form create.
8. Generate a structured report containing the original word, suffix, grammatical category, spelling transformation, extracted base, and normalized representation.
9. Stop the program.

6. Python Program

```
words = ["create", "creates", "creating"]

def analyze_word(word):
    if word == "create":
        return {
            "suffix": "None",
            "category": "Base form",
            "root": "create",
            "transformation": "No inflectional change"
```

```
        }
    elif word == "creates":
        return {
            "suffix": "-s",
            "category": "Third-person singular present",
            "root": word[:-1],
            "transformation": "Remove -s to recover create"
        }
    elif word == "creating":
        return {
            "suffix": "-ing",
            "category": "Present participle",
            "root": word[:-3] + "e",
            "transformation": "Remove -ing and restore final e"
        }
    else:
        return None
print("INFLECTIONAL MORPHOLOGY AND NORMALIZATION REPORT")
print("=" * 60)
for word in words:
    result = analyze_word(word)
    print("\nOriginal Word:", word)
```

print("Identified Suffix:", result["suffix"])

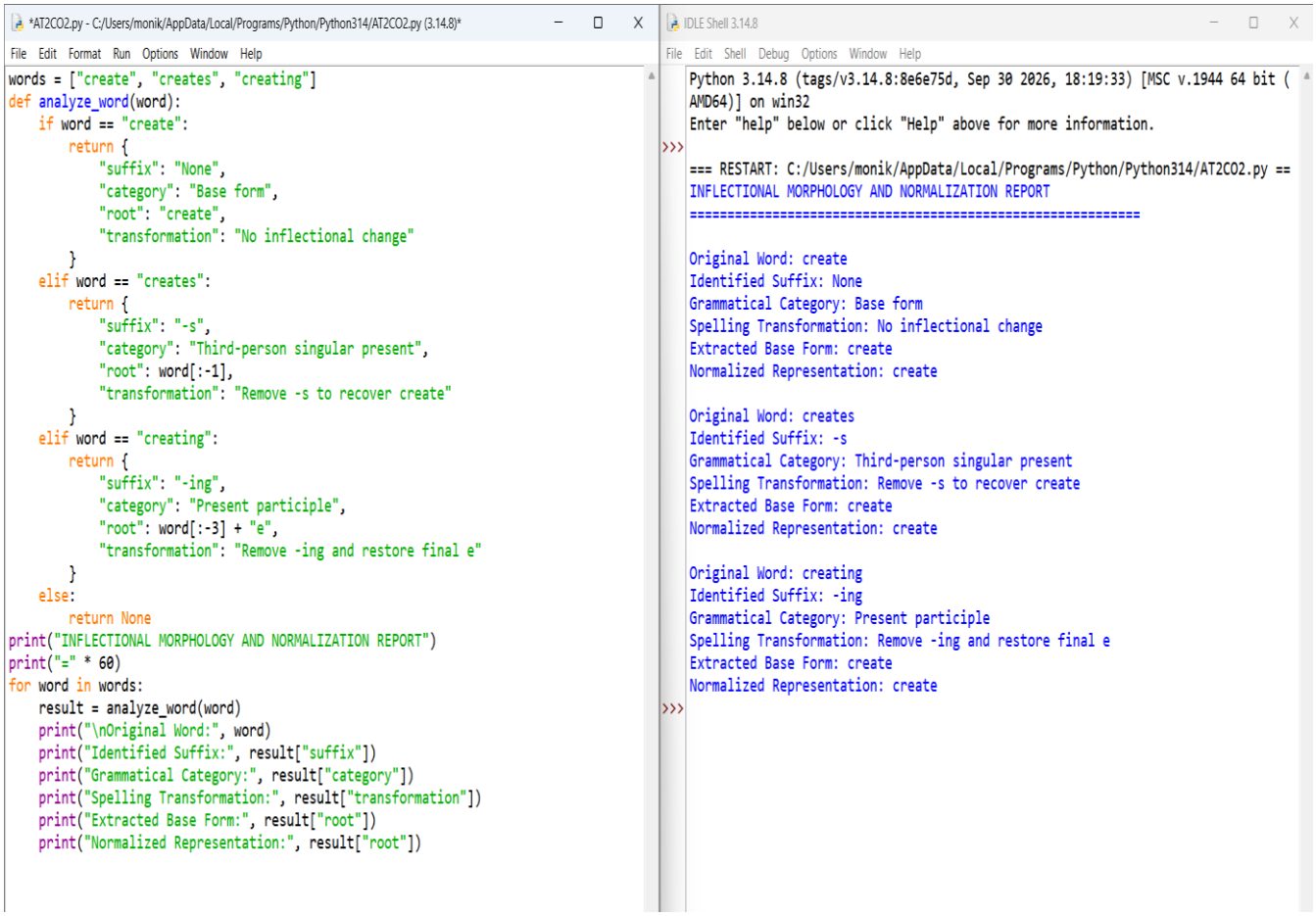
print("Grammatical Category:", result["category"])

print("Spelling Transformation:", result["transformation"])

print("Extracted Base Form:", result["root"])

print("Normalized Representation:", result["root"])

7. OUTPUT



8. Result

The Python program successfully identifies the inflectional forms of create, creates, and creating, classifies their grammatical features, extracts the base form, and normalizes all three words to create. This supports consistent indexing and efficient information retrieval in search systems.